Legal, Social, Ethical & Professional Issues in Programming

CPT111 Java Programming

Lecture 14

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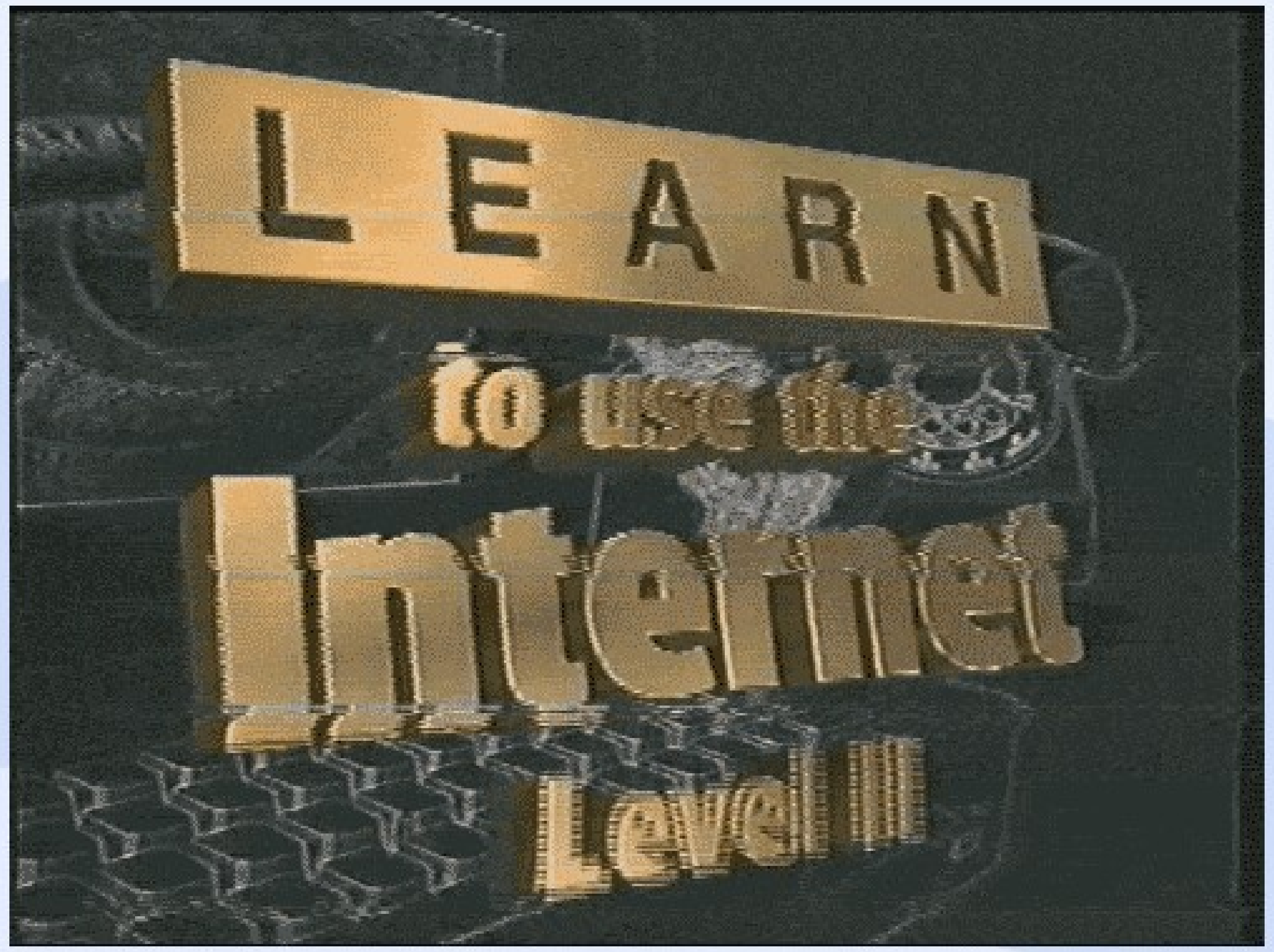
Learning outcomes

→ Explain the Data Protection and data sharing

→ Analyze real world cases



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UK Data Protection Act – info about you

- the General Data Protection Regulation (GDPR), current version 2018
- "controls how your personal information is used by organisations, businesses or the government"
- Your rights
- " find out what information the government and other organizations store about you", including
- Access/update/delete your personal data
- Control the processing of your personal data

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UK Data Protection Act

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Legal protection on your data, regarding

- race
- ethnic background
- political opinions
- religious beliefs
- trade union membership
- genetics
- biometrics (where used for identification)
- health
- sex life or orientation

GDPR personal data can be...

- Name
- Identification number
- Location data
- Physical address
- Email address
- Photograph
- Video
- Voice recording
- Biometric data
- Online identifier

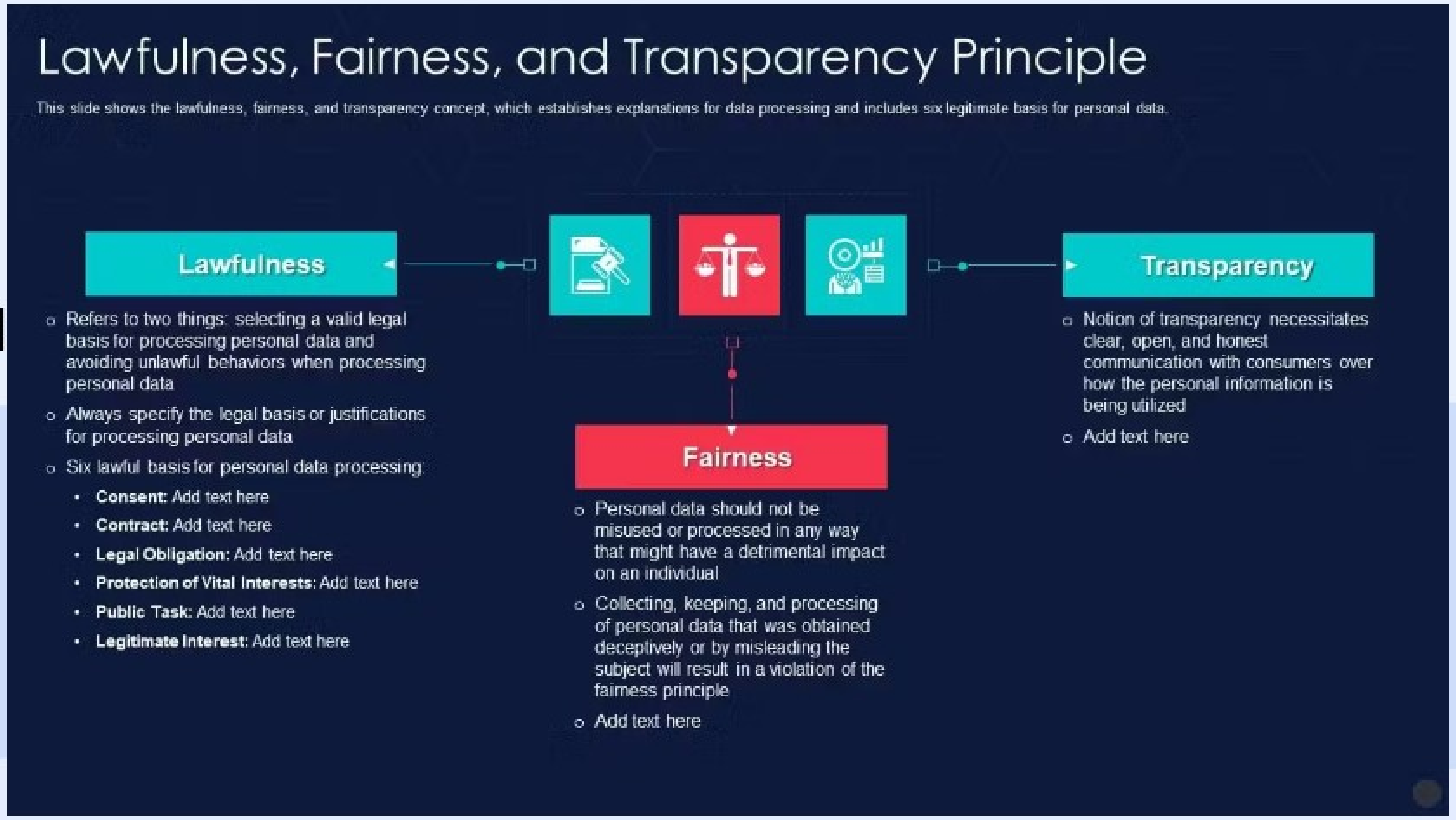
UK Data Protection Act – Principles

- "used fairly, lawfully and transparently"
- "used for specified, explicit purposes"
- "used in a way that is adequate, relevant and limited to only what is necessary"
- "accurate and, where necessary, kept up to date"
- "kept for no longer than is necessary"
- "handled in a way that ensures appropriate security, including protection against unlawful or un-authorised processing, access, loss, destruction or d
- “The controller shall be responsible for, and be able to demonstrate compliance.” - ADDED in the new version

UK Data Protection Act – principles

Lawfulness, fairness and transparency

- Lawfulness means the user has given you consent to do so; you must do it to make good on a contract, and it’s necessary to fulfill a legal obligation.
- Fairness means you won’t mishandle or misuse the data you collect.
- Transparency: being clear, open, and honest with data subjects about who you are, and why and how you’re processing their personal data



Purpose limitation

- This purpose limitation means data is "collected for specified, explicit, and legitimate purposes" only.
- Your purposes for processing data must be clearly established. And they must also be clearly communicated to individuals through a privacy notice.
- Finally, you must follow them closely, limiting the processing of data to only the purposes you've stated.
- What if you want to use the data you've collected for a new purpose that's incompatible with your original purpose?
- Ask specifically for consent again to do it — unless you have a clear obligation or function set out in law.

> (b) collected for specified, explicit and legitimate purposes and not further processed in a manner that is incompatible with those purposes; further processing for archiving purposes in the public interest, scientific or historical research purposes or statistical purposes shall, in accordance with Article 89(1), not be considered to be incompatible with the initial purposes ('**purpose limitation**');
>
> -GDPR, Article 5 (1) (b)

Data minimization

- Only collect the smallest amount of data you'll need to complete your purposes.
- For example, if you want to gather subscribers for your email newsletter, you should only ask for information necessary to send out the newsletters.
- Avoid gathering personal data such as phone numbers or home addresses, which aren't directly related to your purpose.

Accuracy

- It’s up to you to ensure the accuracy of the data you collect and store.
- Set up checks and balances to correct, update, or erase incorrect or incomplete data that comes in.
- Also have regular audits on the calendar to double check the cleanliness of stored data.

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Accuracy

- Controllers are responsible for taking all possible steps to ensure that personal data is accurate because there are obvious risks to data subjects if inaccurate data is processed.
- Personal data must be kept complete and up-to-date data.
- E.g.: Refresh data by sending email to ask customer to update records / remind employees to log in to review records such as emergency contact information.

Storage limitation

- According to the GDPR, you have to justify the length of time you're keeping each piece of data you store.
- Data retention periods are a good thing to establish to meet this storage limitation policy.
- Create a standard time period after which you'll anonymize any data you're not actively using. Once you are done using personal data, delete it immedi

Principle 5: Storage limitation

- Personal data must be kept in a form that makes it possible to identify data subjects for no longer than is necessary for the purposes of the processing.
- Storing these data for longer periods is allowed when the processing of the data will aim at achieving purposes in the public interest, scientific or historical research purposes or statistical purposes.
- Legal requirements for retention can deem data retention necessary, such as record keeping requirements imposed by a regulator.
- Note that where an organisation may need to produce evidence for a legal defence in the future, this may also be a legitimate reason to retain data that is no longer needed for processing.

Integrity and confidentiality

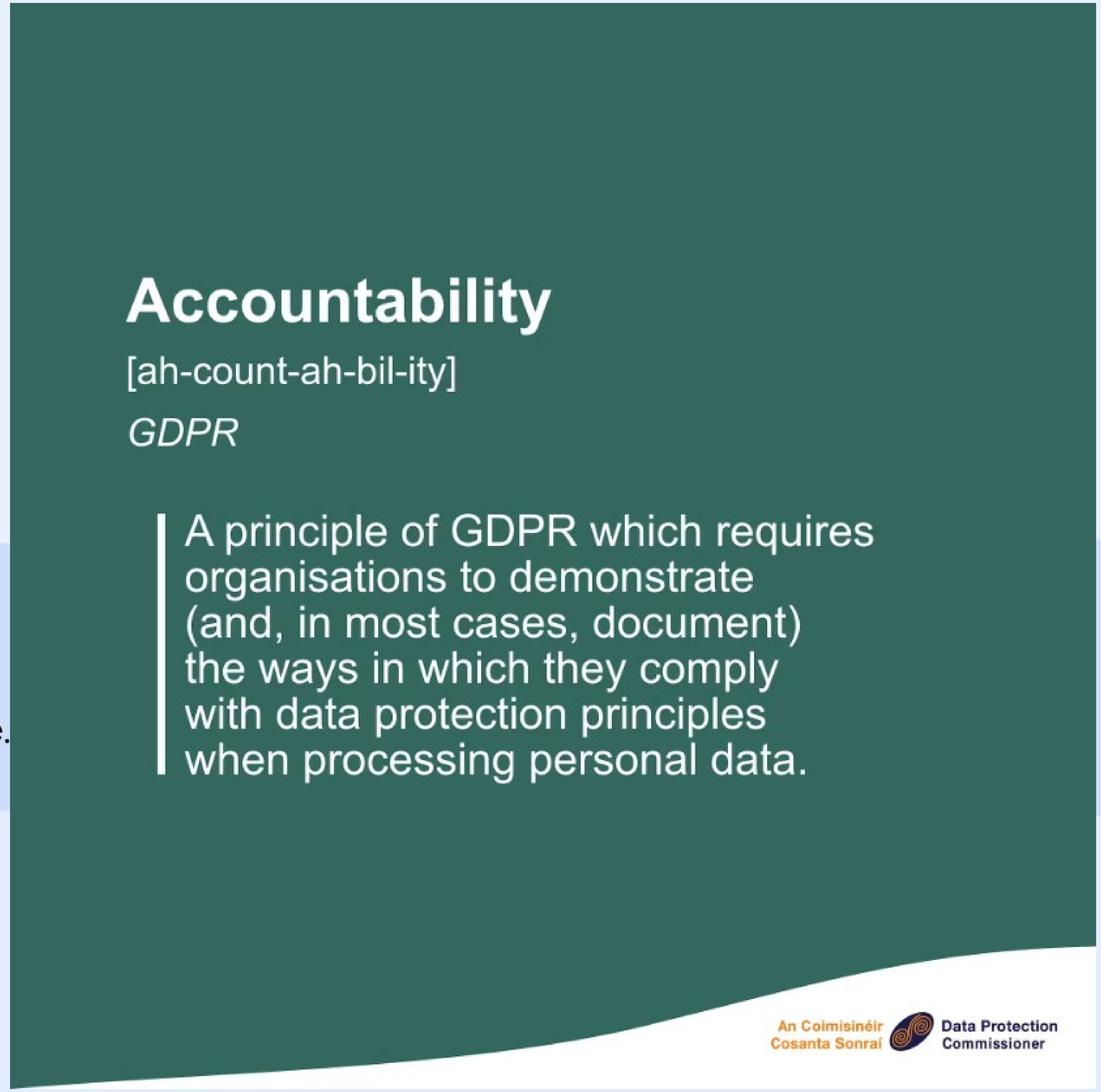
- Maintain the integrity and confidentiality of the data you collect, essentially keeping it secure from internal or external threat
- This takes planning and proactive diligence. Must protect data from unauthorized or unlawful processing and accidental loss, destruction, or damage

Integrity & confidentiality

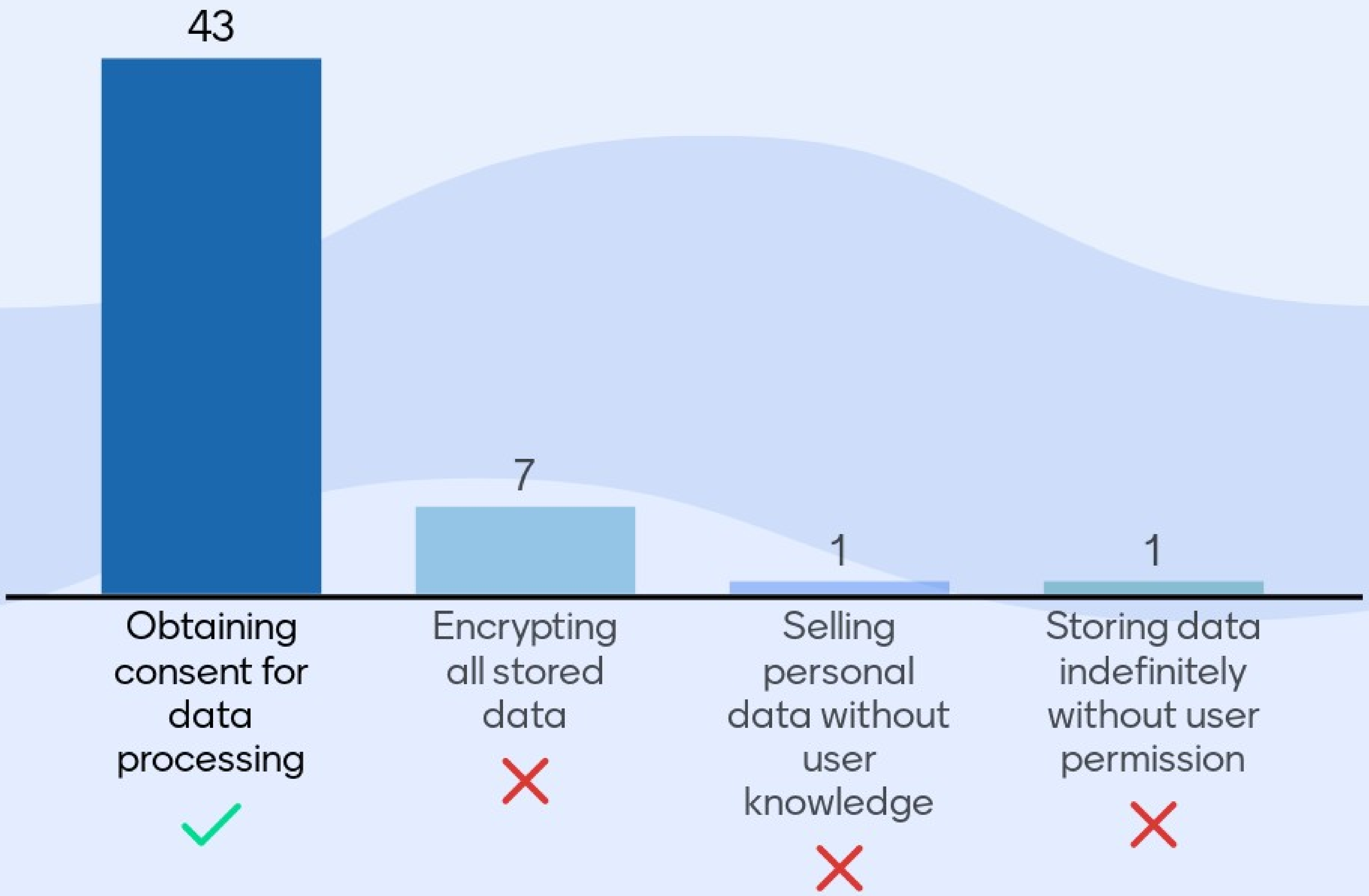
Requires processors to handle data "in a manner [ensuring] appropriate security of the personal data including protection against unlawful processing or accidental loss, destruction or damage".

Accountability

- The GDPR regulators know an organization can say they're following all the rules without actually doing it.
- That's why they require a level of accountability
- You must have appropriate measures and records in place as proof of your compliance with the data processing principles.
- Supervisory authorities can ask for this evidence at any time. Documentation is key here. It creates an audit trail you — and authorities — can follow



Which of the following is a key aspect of the GDPR principle of "lawfulness, fairness, and transparency"?



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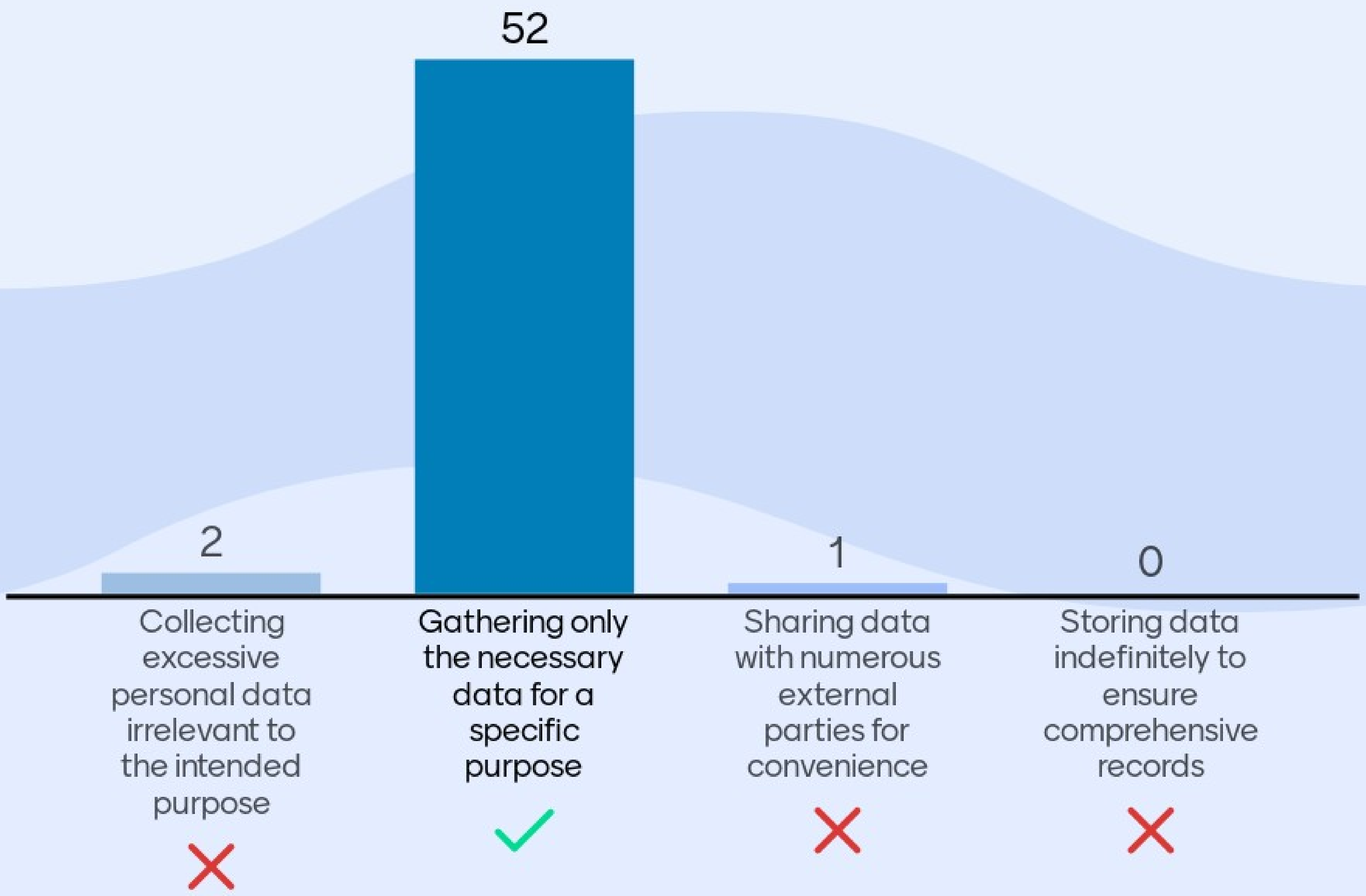
Which action aligns with the GDPR principle of "data minimization"?



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How does the GDPR's principle of "storage limitation" impact data handling?

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UK Data Protection Act – Individual Rights

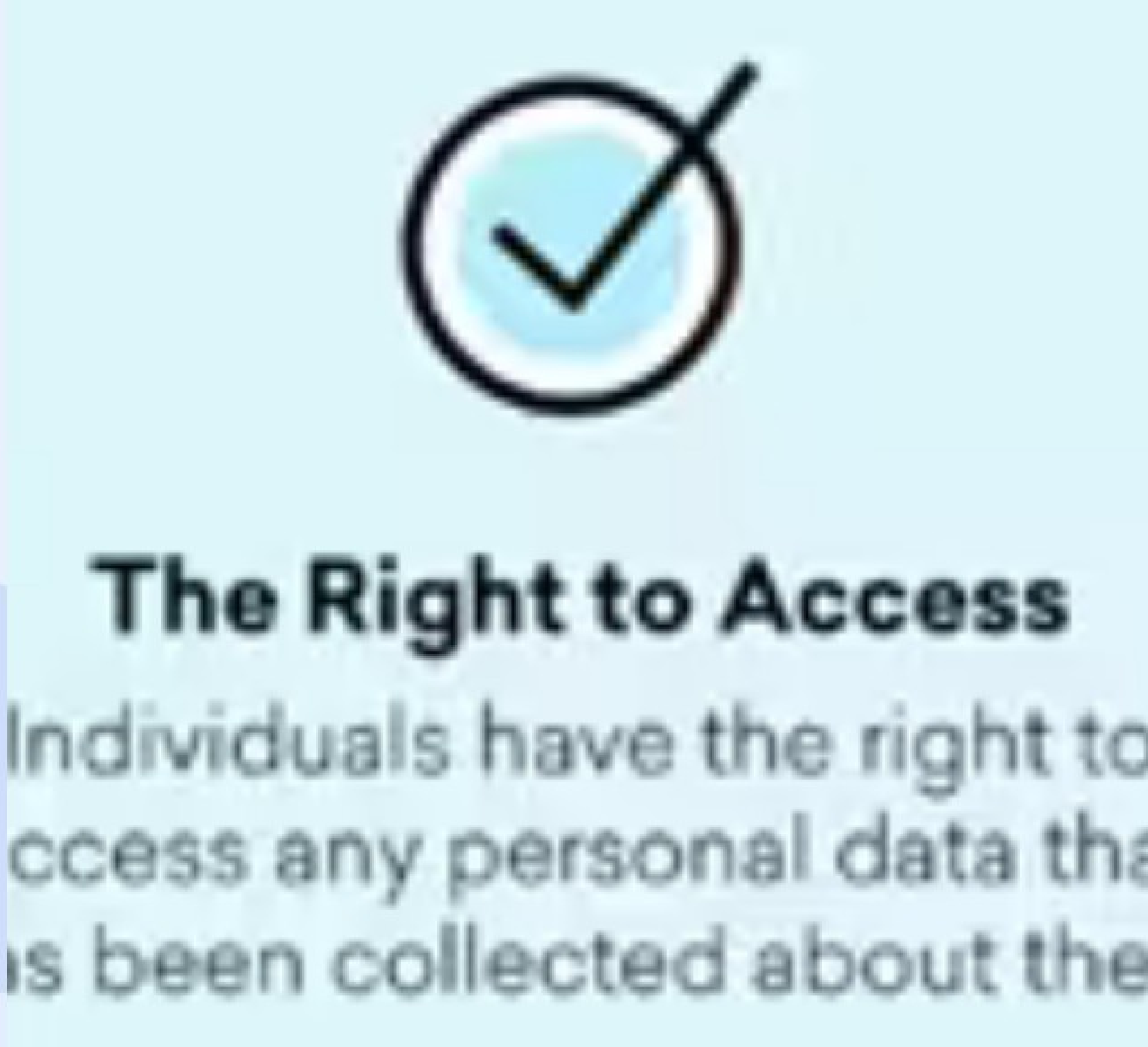
- The right to be informed
- The right of access
- The right to rectification
- The right to erasure
- The right to restrict processing
- The right to data portability
- The right to object
- Rights in relation to automated decision making and profiling.

The right to be informed

- Under the GDPR, individuals have the right to be informed about how companies
- collect and use their personal data, how long they plan to keep that data, and who they'll share it with.
- Companies that collect data have to provide certain information to the data subjects, including the identities and contact details

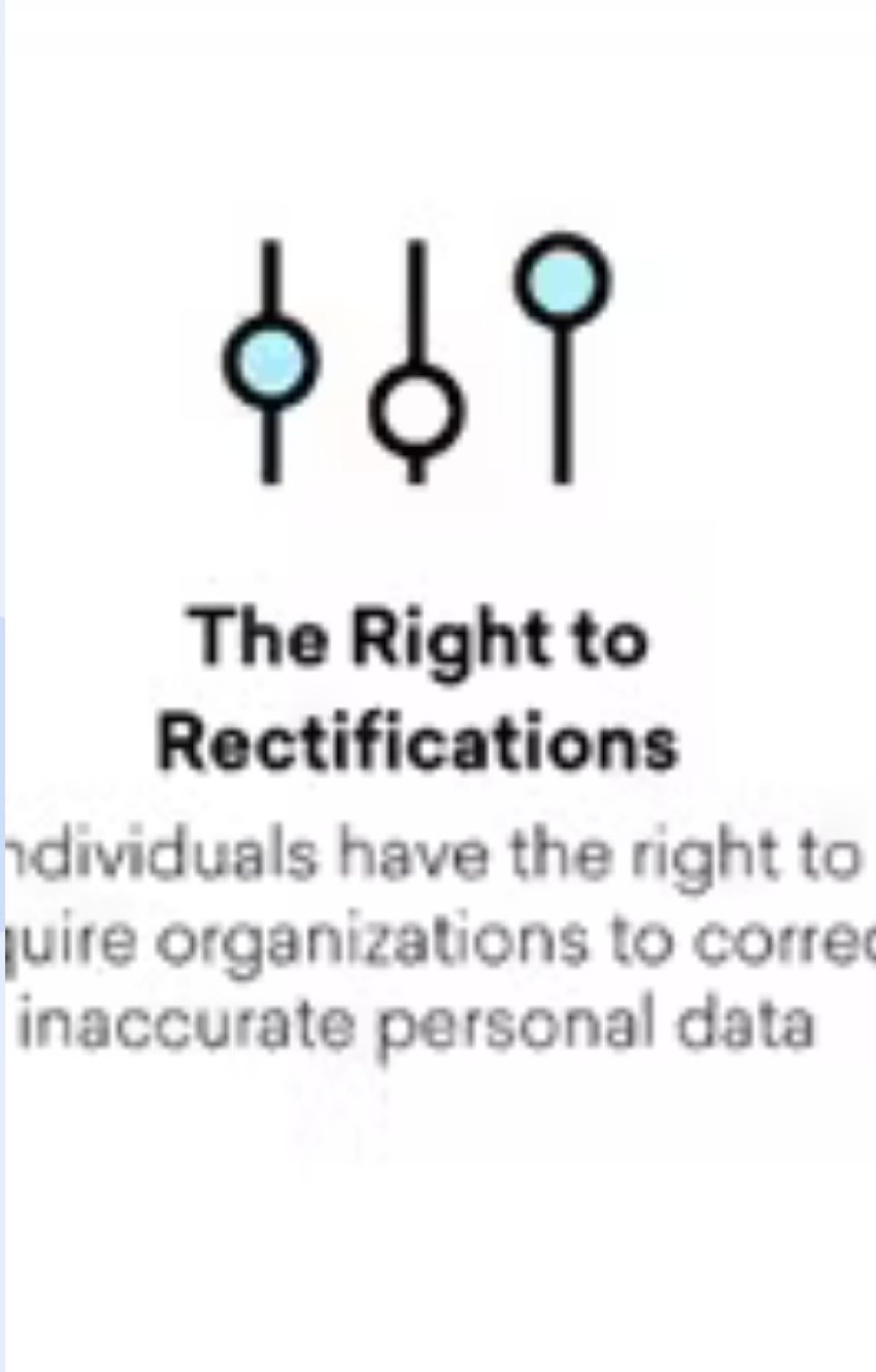
The right of access

- Individuals have the right to know and review exactly what information companies
- have collected, how they're storing and processing that data, and what they're going to do with it.



The right to rectification (correction)

→ Data subjects have the right to have incomplete data completed and incorrect data corrected.



The right to erasure

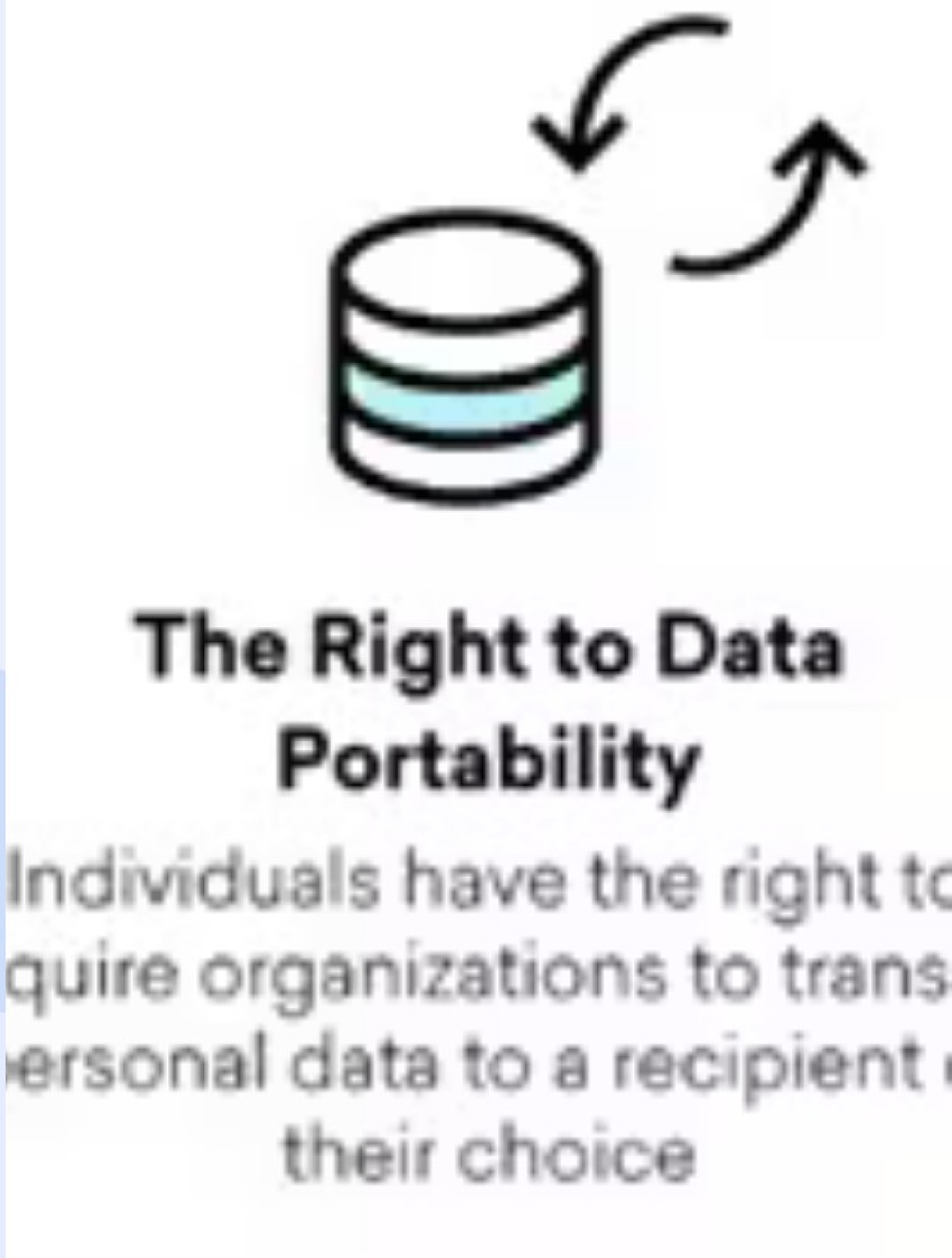
- Data subjects have the right to have personal data permanently deleted. This is also known as the "right to be forgotten."
- In this case, companies can't argue that their legitimate interests in processing users' data outweigh the individuals' rights to have it erased.
- However, this right doesn't apply if the processing of data that's subject to an erasure request is necessary to comply with legal obligations

The right to restrict processing

- If data subjects can't require that data controllers erase their personal information
- they can restrict the ability of data controllers to process that data, under certain circumstances

The right to data portability

- Individuals have the right to obtain and reuse their personal data for their own purposes across different services.
- Data subjects can request that data controllers send their personal data files electronically to third parties.
- If technically feasible, companies must provide the data in commonly used, machine-readable formats.



The right to object

- Data subjects have the right to object to the processing of their personal data in certain circumstances.
- However, companies may still process the data to establish or defend legal claims,
- or if they can demonstrate there are legitimate grounds that override individuals' interests and rights.

The right to not be subject to automated decision making

- Individuals have the right to demand human intervention, rather than having important decisions made by algorithms.
- Companies are required to inform people that they will be subject to algorithmic decision-making and let them know that they can opt out of it.

Ethical and professional issues

- The word: " a set of moral principles : a theory or system of moral values"
- About good and bad; right and wrong
- In theory: moral philosophy
- Applications: war, environment, animal rights, research

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Law vs. Ethics

- Law alone can't restrict human behavior
- Impractical/impossible to describe/enforce all acceptable behaviors
- Ethics/morals are sufficient self-controls for most people
- Ethics is not religion (but religions include ethical principles

Law vs. Ethics

- Ethical principles are not universal
- Vary in different cultures
- Vary even in different individuals in the same culture
-
- Ethics is pluralistic in nature
- In sharp contrast to science and technology that often has only one correct answer

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What is Ethics?

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Ethical Principles

- Golden Rule: "Do unto others as you would have them do unto you" (Luke 6:31)
- in Chinese "己所不欲，勿施于人" 《论语》
- Immanuel Kant's Categorical Imperative: "If an action is not right for everyone to take, then it is not right for anyone "
- Descartes' rule of change: "If an action cannot be taken repeatedly, then it is not right to be taken at any time"

Ethical Principles

- Utilitarian Principle: "Take the action that achieves the greatest value for all concerned"
-
- Risk Aversion Principle: "Take the action that produces the least harm or incurs the least cost to all concerned "
-
- Ethical no-free-lunch rule: "Assume that all tangible and intangible objects are owned by someone else, unless shown the contrary." " If someone has c

Academic Ethics Code of Conduct

- Plagiarism, Cheating, and Fabrication of Data.
- XJTLU e-Bridge - >
- Student Academic Services/ Quick reference ->
- Policies and Regulations ->
- Academic Integrity Policy
- POLICY ON STUDENT CONDUCT AND DISCIPLINE

Academic Ethics Code of Conduct

- Offences can result in attendance at a University-level committee and penalties being imposed.
- Prevention: Turnitin, citation, quotation ""

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Protecting Programs and Data - IP

- Legal protections:
- copyrights, trademarks, trade secrets, and patents
- Categories:
- Inventions, trademarks, logos, and industrial designs
- Copyrighted material (Literary and artistic works, online materials
- Fair Use Doctrine: for certain purposes

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Protecting Programs and Data

- Copyrights — designed to protect expression of ideas (creative works of the mind)
- Ideas themselves are free
- Different people can have the same idea
- The way of expressing ideas is copyrighted
- Copyrights are exclusive rights to making copies of expression

Copyright Rules to Remember

1. Just because you found it online, doesn't mean it's free to use (even if you're a teacher or student).
2. There are a lot of resources you can use freely including work that has a Creative Commons license or is in the public domain.
3. You have a right as a creator to have your work protected from copying and you can also give your own content a Creative Commons license.
4. If in doubt about using content, ask the creator for permission, find a free alternative, purchase an alternative, or make your own material.
5. Instead of looking for loopholes, consider whether you're being the most responsible and ethical digital citizen you can be.

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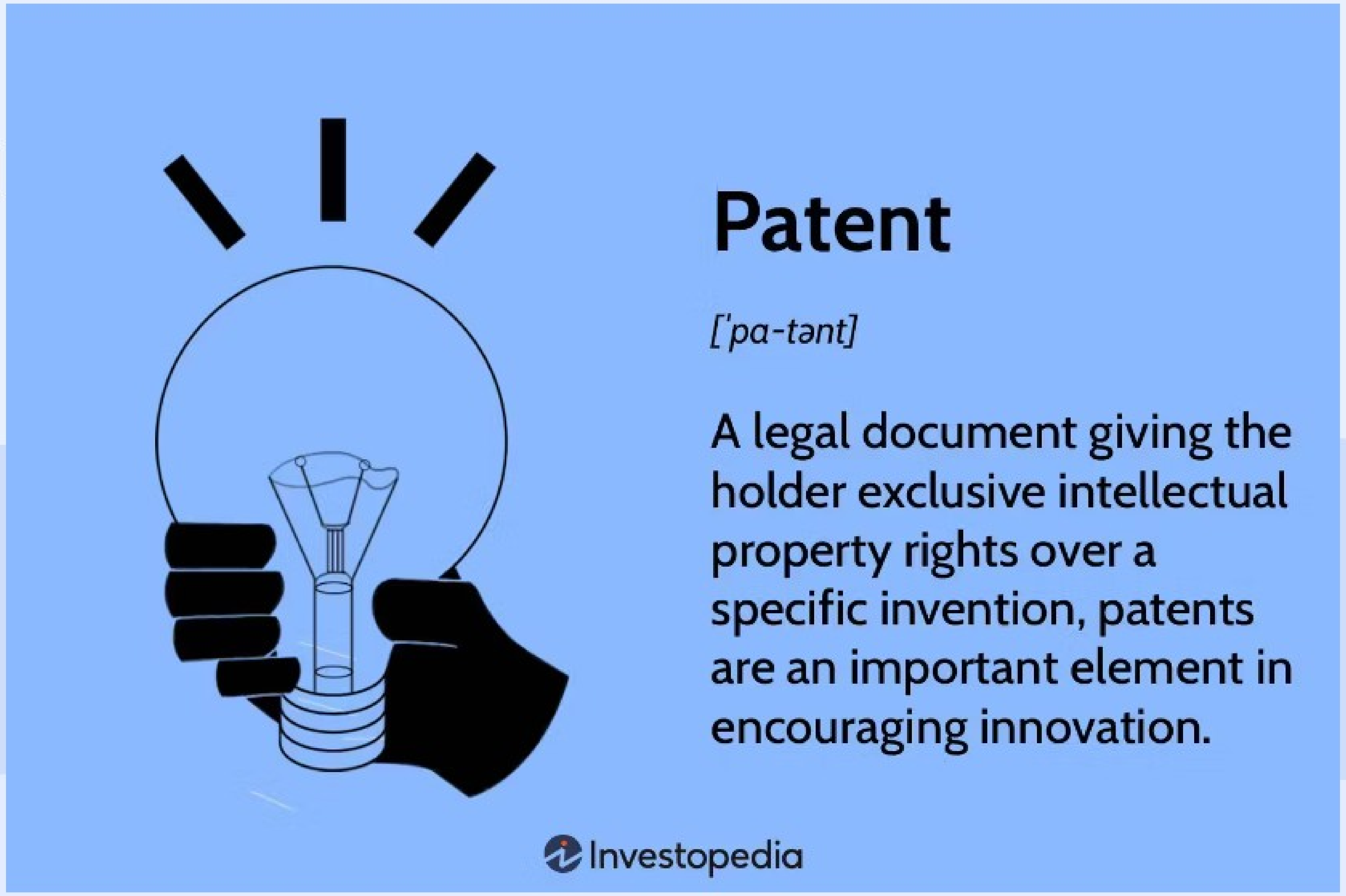
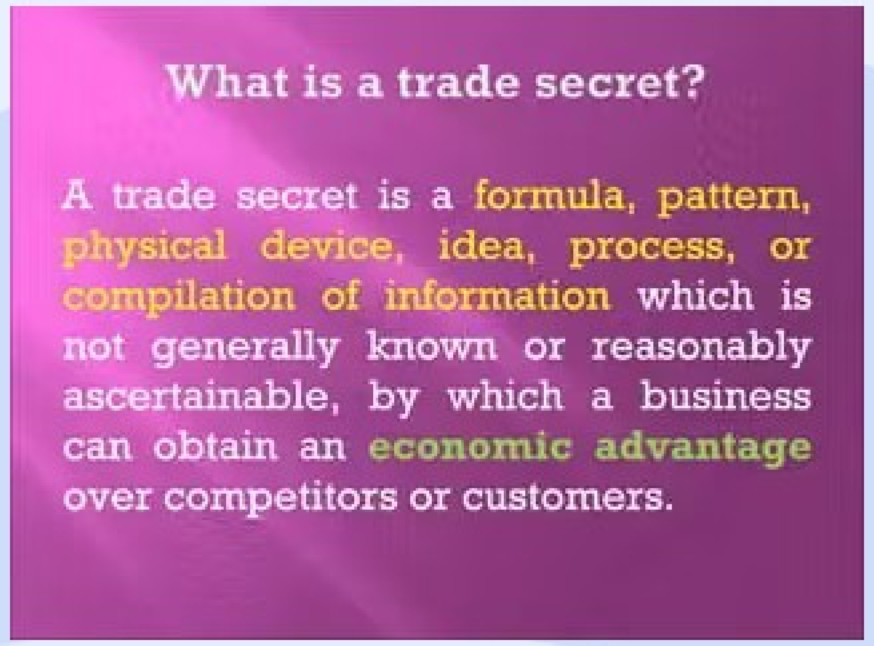
Protecting Programs and Data

- Patent — designed to protect tangible objects, or ways to make them (not works of the mind)
- Protected entity must be novel & nonobvious
- The first inventor who obtains patent gest his invention protected against patent infrigement
- Patents applied for algorithms only since 1981



Protecting Programs and Data

- Trade secret — information that provides competitive edge over others
- Information that has value only if kept secret
- Undoing release of a secret is impossible or very difficult
- Reverse engineering used to uncover trade secret is legal!
- T.s. protection applies very well to computer s/w
- E.g., pgms that use algorithms unknown to others



	Copyright	Patent	Trade Secret
Protects	Expression of idea, not idea itself	Invention—way something works	Secret, competitive advantage
Protected Object Made Public	Yes; intention is to promote publication	Design filed at Patent Office	No
Must Distribute	Yes	No	No
Ease of filing	Very easy, do-it-yourself	Very complicated; specialist lawyer suggested	No filing
Duration	Originator's life + 70 yrs; 95 y. For company	19 years	Indefinite
Legal Protection	Sue if unauthorized copy sold	Sue if invention copied/reinvented	Sue if secret improperly obtained

Comparing Copyright, Patent and Trade Secret Protection

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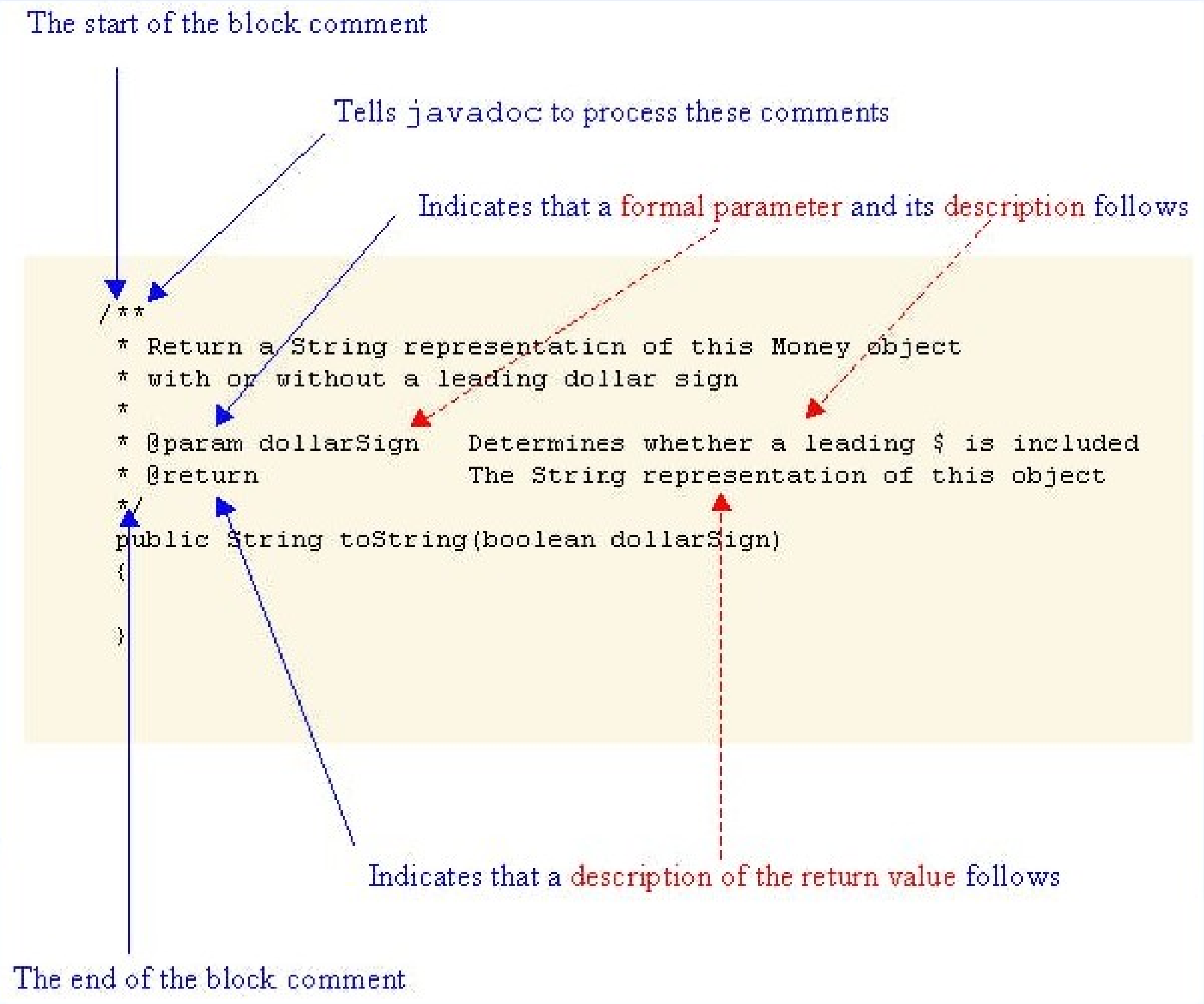
Structure of a Javadoc comment

- A Javadoc comment is set off from code by standard multi-line comment tags / *and* /. The opening tag (called begin-comment delimiter), has an extra a
- The first paragraph is a description of the method documented.
- Following the description are a varying number of descriptive tags, signifying:
- The parameters of the method (@param)
- What the method returns (@return)
- Any exceptions the method may throw (@throws)
- Other less-common tags such as @see (a "see also" tag)



```java
// import statements

/**
 * @author      Firstname Lastname <address @ example.com>
 * @version     1.6                   (current version number of program)
 * @since       1.2           (the version of the package this class was first added to)
 */
public class Test {
    // class body
}
```

Class

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```
/**
 * Short one line description.                              (1)
 * <p>
 * Longer description. If there were any, it would be      (2)
 * here.
 * <p>
 * And even more explanations to follow in consecutive
 * paragraphs separated by HTML paragraph breaks.
 *
 * @param  variable Description text text text.            (3)
 * @return Description text text text.
 */
public int methodName (...) {
    // method body with a return statement
}
```

Method

The last...
Please complete the MQ on the course page NOW!

– Teng Ma and Erick